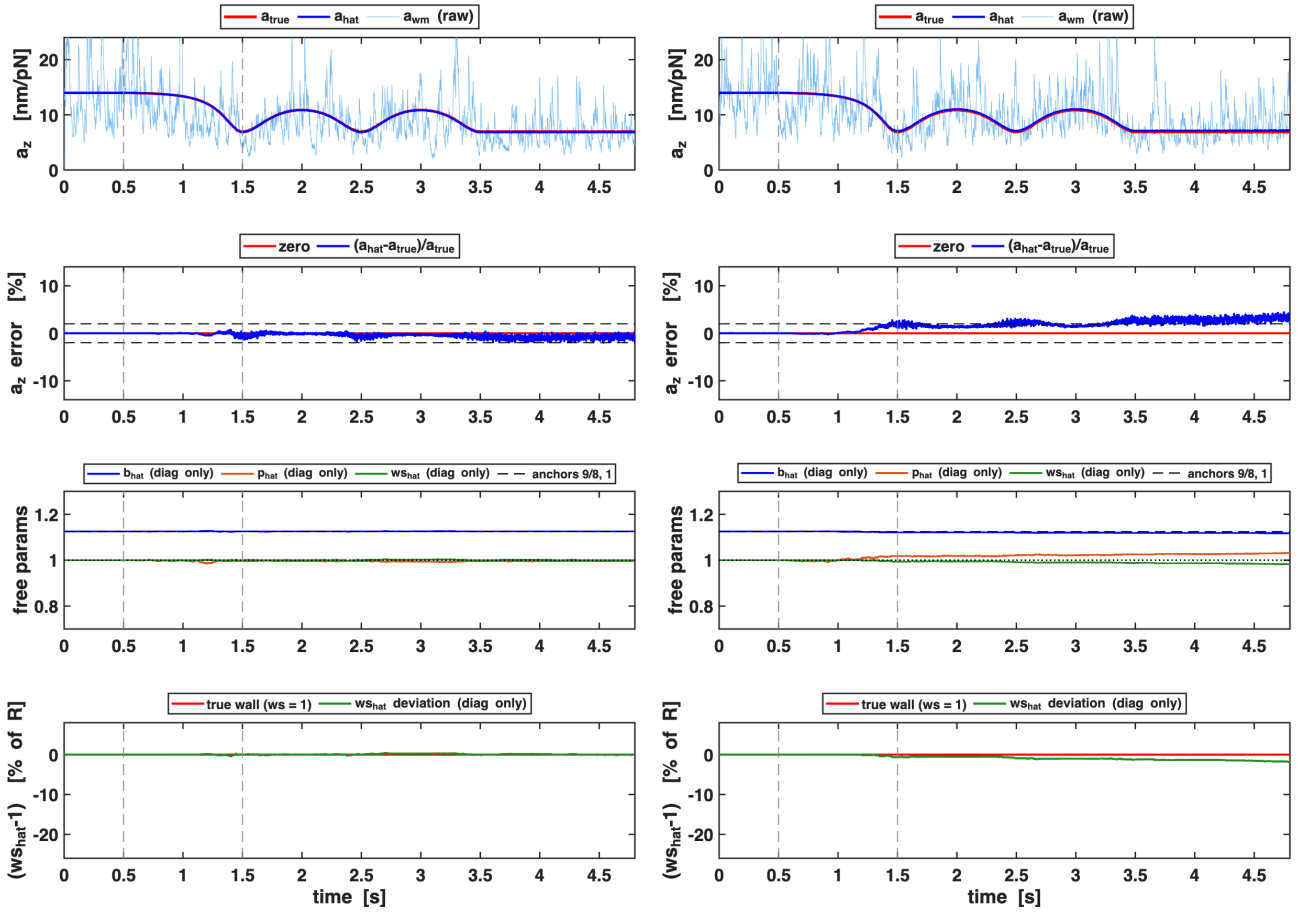
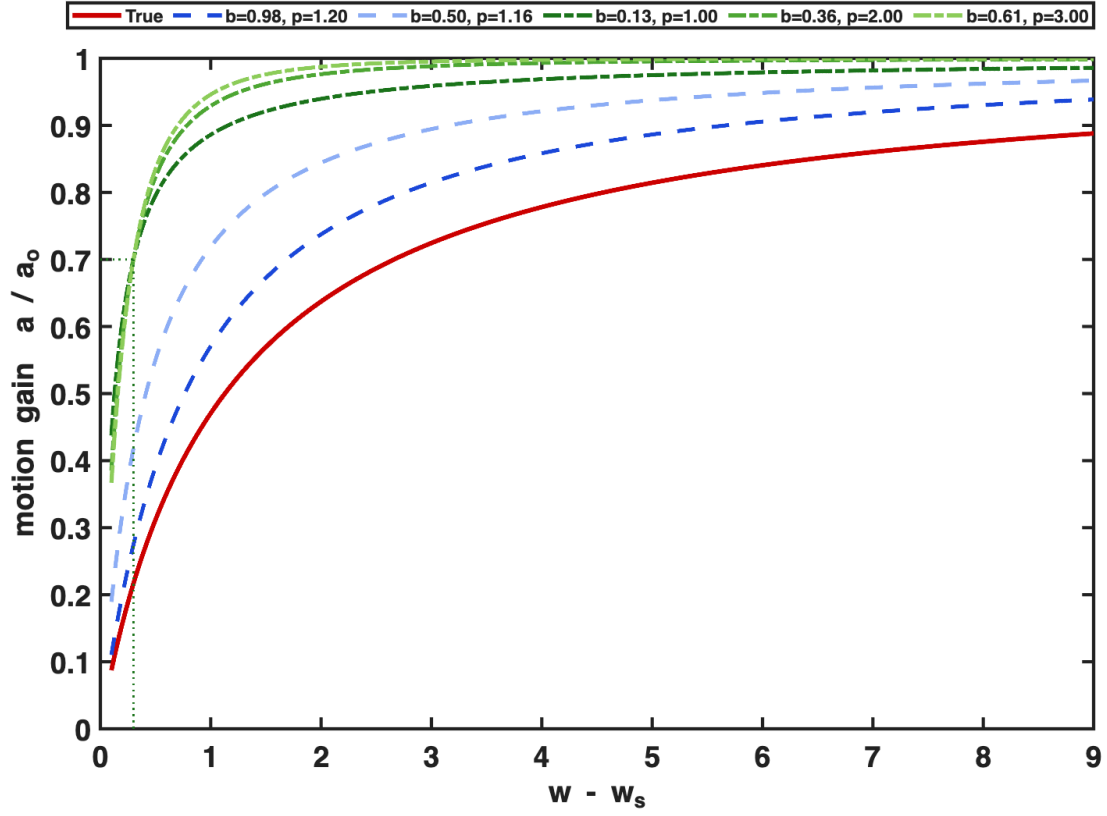


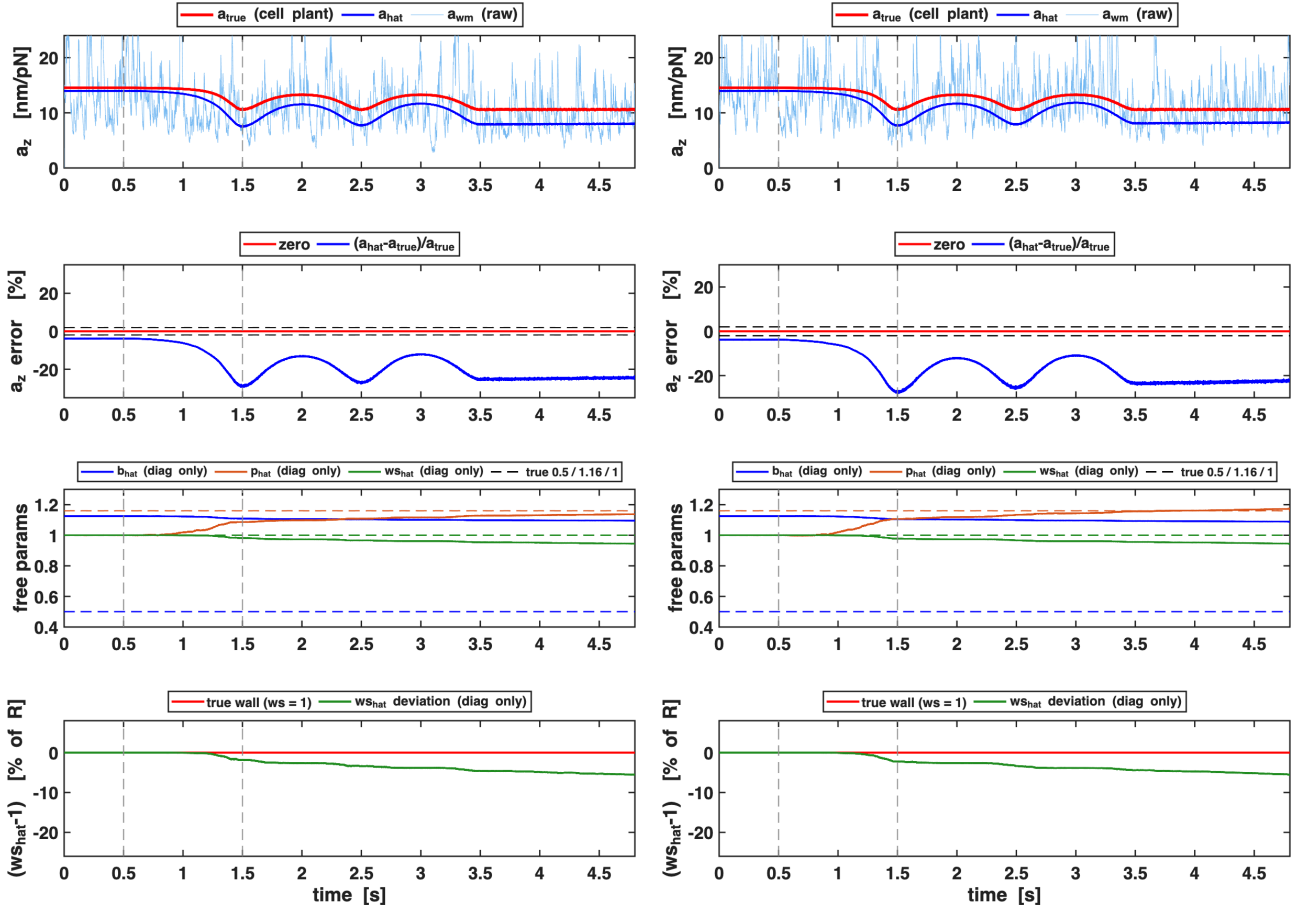
Selected figures (2026-08-03)



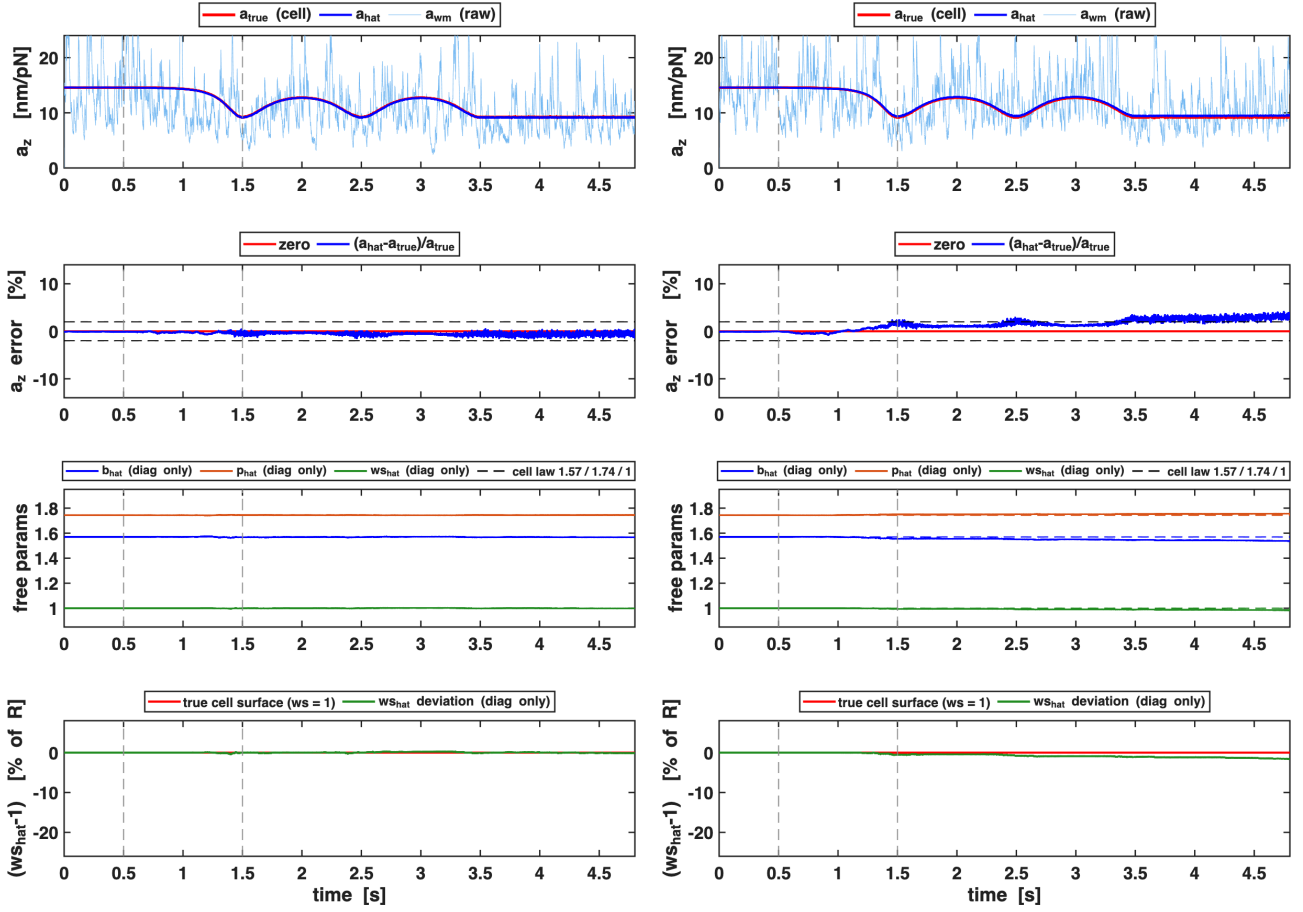
Config 3 — estimate (b, p, w_s) with the calibrated-grade prior $\sqrt{P_{w_s}}[0] = 0.028$ (house 0.111 elsewhere unchanged): seed 7 (left) vs seed 11 (right); the seed-11 $\hat{w}_s$ end deviation shrinks to -1.8% and the residual realization exits through the b, p channels.



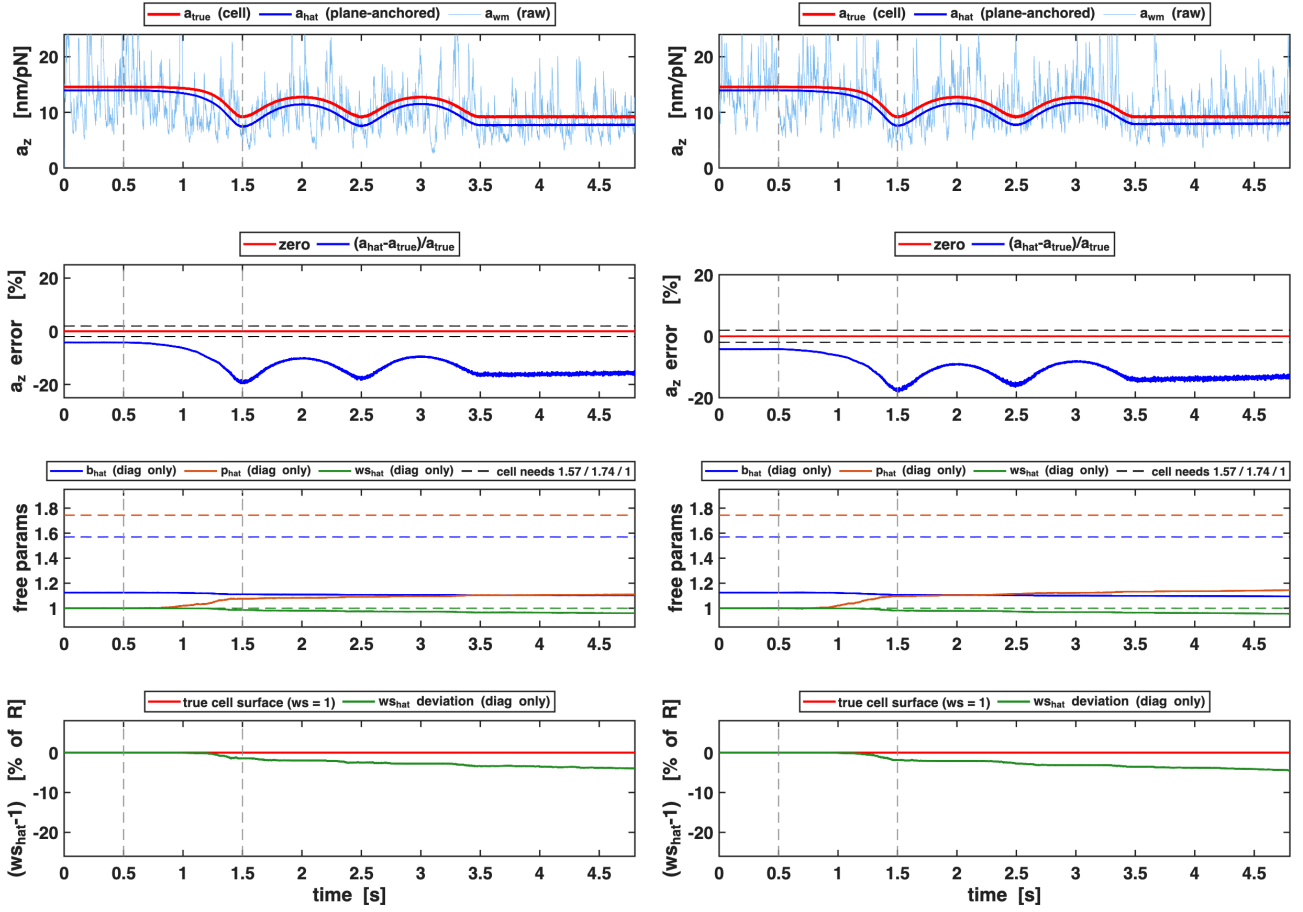
Form B refit to two cell (two-sphere) truths against the rigid-plane truth (red): the family absorbs boundary curvature, but the fitted (b, p) leave the plane anchors $(9/8, 1)$. The three green dash-dot curves are added by hand: all pass through $\bar{a} = 0.70$ at a gap of $0.3 R$ (the rigid plane is at 0.21 there) with the origin still on the surface, and they show the one-parameter freedom a single point leaves — p sets the far-field decay $1 - \bar{a} \sim g^{-p}$ and b follows as $b = rG/(1 - r)$, $r = 0.3^{1/p}$.



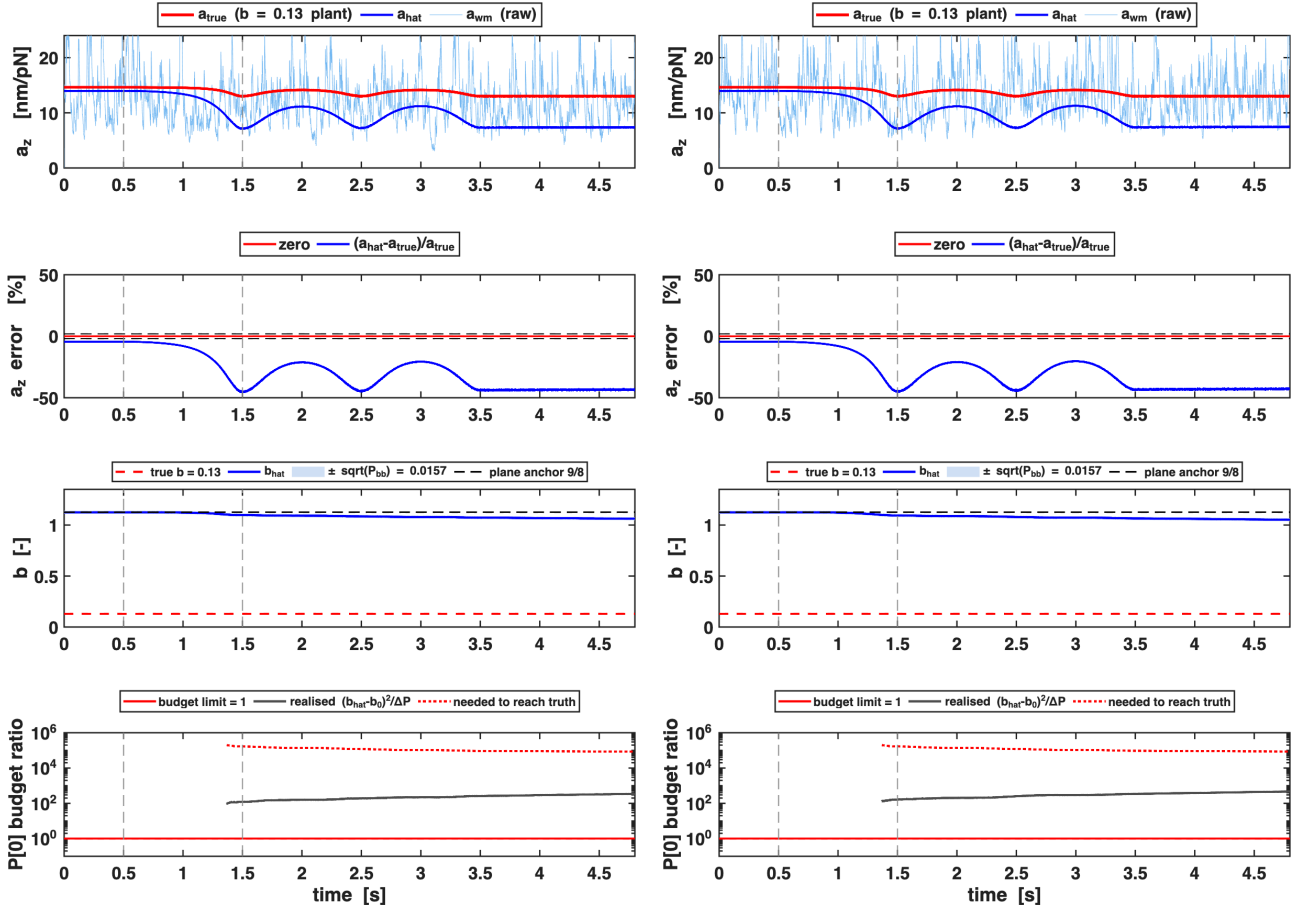
Curve-mismatch arm — same config 3 / prior 0.028 setup, but the TRUE z curve is the cell fit $(b, p, w_s) = (0.5, 1.16, 1)$ while the controller keeps every plane-derived prior: gain error blows up (descent peak $\approx 28\%$, hold $\approx -24\%$), the b, p traversal budgets fail by $\sim 50\times$, and $\hat{w}_s$ is dragged -5.5% against its own 2.8% prior — the plane-anchored prior widths cannot absorb a boundary this far outside the family they were derived for.



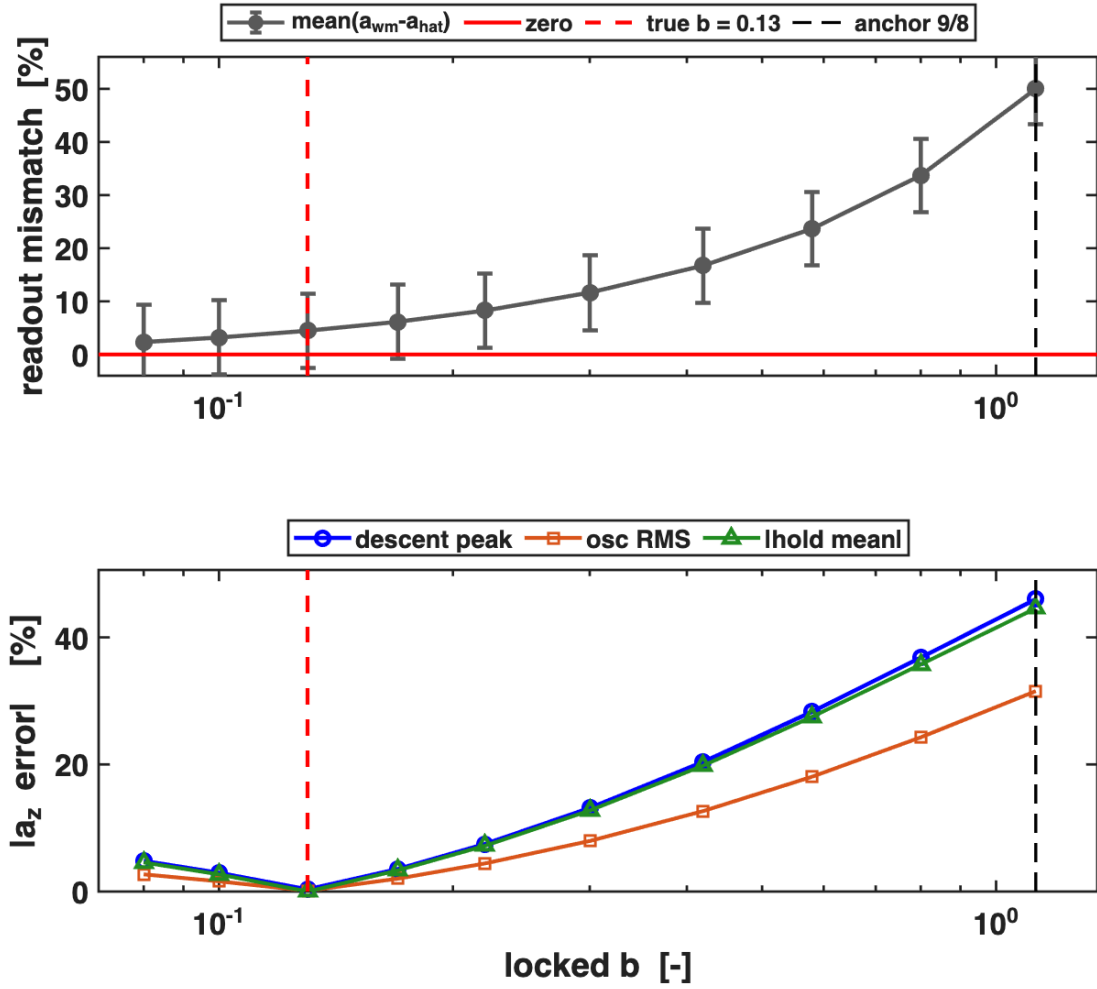
Cell boundary done properly — plant is the exact two-sphere truth for $R_c = 5 \mu\text{m}$ and the controller carries the cell law $(b, p, w_{s0}) = (1.570, 1.744, 0.809)$ with widths $[0.042, 0.022, 0.015]$, fitted offline to that same published curve (sup 0.086 %): performance returns to the plane baseline (descent 1.10/2.33 %, hold $-0.61/+2.82$ %, $\hat{w}_s$ ending $-0.1/-1.6$ %), so the difficulty was never the curvature but the anchor. This page is the capability bound (“if we knew”), not the operating case.



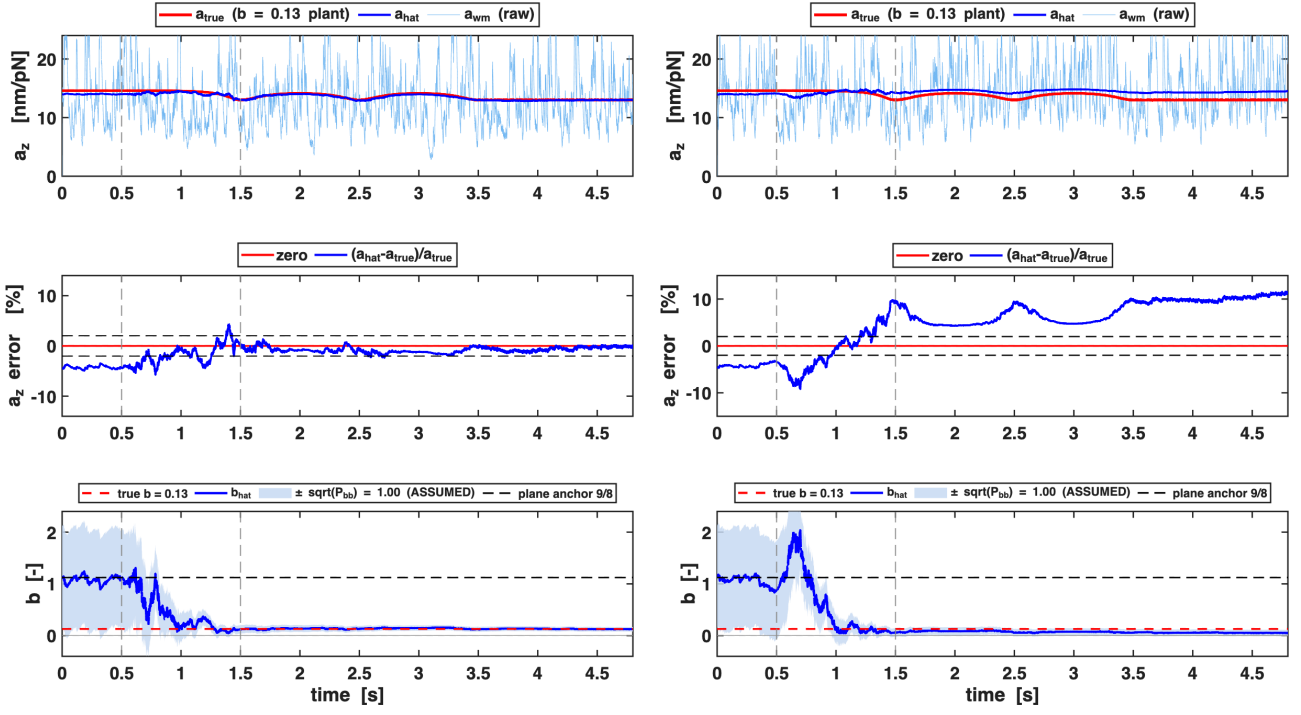
The operating case — same $R_c = 5 \mu\text{m}$ cell plant, but the controller knows only the plane (anchors 9/8, 1, priors 0.0157/0.0245, origin at the nominal contact): descent peak 19.6/17.8 %, hold $-16.0/ -13.6$ %, b, p stall near their plane anchors (final 1.10/1.11 against the 1.570/1.744 the cell demands), $\hat{w}_s$ absorbs part of the mismatch by drifting $-4.0/ -4.5$ %, and the traversal budgets fail by $25\text{--}40\times$ — a plane-anchored filter meeting a cell is wrong by ~ 15 % and says so, but cannot repair itself.



Single-free-parameter arm (2026-08-05) — the TRUE z curve is the $p = 1$ member of the page-2 constructed family, $(b, p, w_s) = (0.13, 1, 1)$, and the controller frees ONLY b : p and w_s are locked *at their true values*. Nothing is left to blame — no representation error (the truth is a member of the controller’s own family), no b - p coupling, no b - w_s degeneracy — so the only remaining obstacle is the prior width itself. The envelope-derived $\sqrt{P_{bb}}[0] = 0.0157$ places the truth 63.5 prior widths from the 9/8 seed; $\hat{b}$ moves 0.064/0.073 of the 0.995 it would need and stalls at 1.061/1.052, the traversal budget realises 348/472 against a limit of 1 while reaching the truth would require $\sim 8.5 \times 10^4$, and the gain error settles at descent peak 45.5/45.2 %, hold $-43.5/-43.0$ %. The plant here is a *constructed* test curve, not a fitted cell (the physical cell fits in this document sit at $b = 0.50$ – 1.57 , depending on whether the law origin is held on the surface): the page therefore indicts the prior structure, not the boundary physics.



Locked- b profile for the same constructed plant — b is LOCKED on a swept grid instead of estimated, so the EKF prior plays no part. Bottom: the truth-referenced gain error is a clean V bottoming *exactly* at $b = 0.13$ (descent 0.33 %, osc 0.12 %, hold +0.01 % — the plane baseline), which verifies the Form B machinery: locked at the right constant the filter is exact, and every percent above that floor is the constant being wrong. Top: the data-only statistic — the mean readout-versus-model mismatch, computable without knowing the truth — climbs monotonically across the grid, +2.3 % at $b = 0.08$ to $+50.0 \pm 6.7$ % at the plane anchor, where it stands 7.5σ from zero on two seeds. The run therefore *does* carry, in a truth-free statistic, the information that the anchored b is wrong; the companion page’s filter simply cannot act on it, because its prior places the truth 63.5 widths away. Identifiability is not the obstacle — the prior width is. Two caveats on reading b off this curve: the mismatch is monotonic with no feature at the truth (locating b needs the readout offset characterised first — it is $+4.5 \pm 7.0$ % there, consistent with zero but unresolved at $N = 2$), and near the truth the slope is only ~ 6 % per e-fold against that 7 % scatter, so two seeds fix b to a factor of a few, not to a percent. The filter’s own whitened y_2 channel is a poor route to the same information: its innovation mean tracks b over $12\times$, but the per-sample RMS is noise-dominated and bottoms at $b = 0.80$.



Widened- b -prior arm — same plant and same locks as two pages back, but $\sqrt{P_{bb}}[0]$ is opened up so the truth is inside the prior. **The widths are ASSUMED, not derived** (provisional, to bracket the travel $|9/8 - 0.13| = 0.995$); a committed width must come from a declared boundary family.

Ladder, $\sqrt{P_{bb}} \rightarrow \hat{b}_{\text{final}}$ (seed 7 / 11), descent error, budget: $0.0157 \rightarrow 1.061/1.052$, 45.5/45.2 %, 348/472; $0.10 \rightarrow 0.316/0.185$, 29.3/24.8 %, 96/136; $0.25 \rightarrow 0.132/0.051$, 8.5/6.9 %, 16.5/19.7; $0.50 \rightarrow 0.118/0.051$, 5.5/9.6 %, 4.1/4.7; $1.00 \rightarrow 0.123/0.051$, 5.7/9.8 %, 1.01/1.16. Four readings. (i)

Once the prior stops forbidding it the filter *does* walk b to the truth, and it does so during the descent, exactly where the wall-gap information lives. (ii) The $P[0]$ budget criterion turns out to mean literally what it says — to move a distance Δ lawfully the prior must satisfy $\sqrt{P[0]} \geq \Delta$, and at $\sqrt{P_{bb}} = 1.00 \approx \Delta$ the realised ratio lands at 1.01/1.16, on the limit. (iii) The prior stops binding above $\sqrt{P_{bb}} \approx 0.25$: from there the answer is data-limited, not prior-limited, and widening further buys nothing. (iv) What data-limited costs: $\hat{b}$ scatters 0.123 vs 0.051 across two seeds (factor 2.4) and the hold error splits -0.5% vs $+10.2\%$, so the gain recovers to the few-percent level but not to the 0.33 % floor that *knowing* b gives. Caveat on the container: a linear Gaussian prior with mean $9/8$ and $\sqrt{P} = 1.0$ puts $\sim 13\%$ of its mass at $b < 0$, which is unphysical for a length — a genuinely wide prior on b probably wants a log parametrisation, which is a state-level change and was not attempted.